

Nathan Taylor

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Experience

2025-curr.	Semgrep Staff Software Engineer, Program Analysis Performance lead for Semgrep's static analysis engine, improving, in total, p95 scan times by over 3x. Reduced the remote MCP server's cold start latency by 75%. Led the migration to Multicore OCaml, unlocking fine-grained parallelism opportunities. Maintained Semgrep's OCaml compiler fork, comprised of backported fixes and custom improvements.	Seattle, WA
2021-2025	The University of Texas Graduate Research Assistant, Department of Computer Science Researched lightweight formal methods, which trade completeness guarantees for applicability, for validation of systems software. Verified crash consistency of a persistent memory filesystem, and mechanised validating an optimised Paxos implementation against its manually-written correctness proof.	Austin, TX
2019-2021	Independent Consultant Took a break from corporate life to teach, read research papers, and help out on interesting problems: - At <i>Microsoft Research</i>, built an experimental storage engine that combines low-latency ML and dynamic analysis to optimise index structure layout. Reduced critical-path inference latency by 40%. Mentored PhD interns. - At <i>RLCore Technologies</i>, built reliable and safe systems for reinforcement learning-driven industrial control systems. Implemented property-based testing and record/replay debugging libraries for RL agents. - At <i>The University of Toronto</i> and <i>MacEwan University</i>, developed and taught undergraduate PL and systems courses.	
2018-2019	Apple Systems Software Engineer Developed high-performance GPU simulators modeling in-development SoCs for power-constrained devices. Trading cycle-accuracy for usability, it reproduced functional behaviour at interactive framerates, allowing developers to start programming before tapeout. Revived the dormant company-wide systems paper reading group.	Cupertino, CA
2017-2018	Fauna Senior Software Engineer Developed Fauna's strongly consistent, distributed database. Built a greybox fault injection system to state correctness properties, catch safety violations, and maintain development velocity. Mentored engineers new to Scala, JVM concurrency, and functional programming.	San Francisco, CA
2014-2017	Fastly Senior Software Engineer Maintained Fastly's reverse HTTP proxy and cache. Designed and built a sandboxing dynamic analysis runtime for the Fastly software stack, atop which the compiler, API, and security teams built custom analyzers.	San Francisco, CA
2012-2014	Twitter Software Engineer II Extended Twitter's Ruby and Java runtimes, improving the GC, JIT, and perf analysis tooling. Collaborated with service owners to diagnose performance issues. Maintained legacy systems and implemented new services for antisпам classification and actioning.	San Francisco, CA

Education

2021-2024	The University of Texas PhD, Computer Science (incomplete) Supervisor: James Bornholt Organised the Systems+PL reading group and the undergraduate systems directed reading group. Interned as an Applied Scientist in the Automated Reasoning and Database Services groups at AWS, working on at-runtime model checking of distributed systems.	Austin, TX
2009-2012	The University of British Columbia M.Sc., Computer Science Supervisor: Andy Warfield Served as president of the CS Graduate Students' Association. Organized TA training and department social events. Appointed to the UBC Graduate Council. Led the systems and security reading seminars. Received a TA award for my work supporting CPSC 110.	Vancouver, BC, Canada
2005-2009	The University of Alberta B.Sc. Specialization, Computing Science Advocated for students' issues on the Undergraduate Association of Computing Science executive. Acted as the computational chemistry expert on the U of A's Cluster Challenge Team.	Edmonton, AB, Canada

Selected Publications and Presentations

	I also maintain a blog about formal methods, programming language theory, and low-level and concurrent systems.	
09.2024	An Invitation to Liquid Types	Papers We Love NYC Video
07.2024	SquirrelFS: Using the Rust Compiler to Check Filesystem Crash Consistency	OSDI '24 PDF
05.2018	The Life of a FaunaDB Query	Guest post on the Fauna blog
11.2016	Hands-on HTTP/2, a Fresh Start to The Web	QCon SF
06.2016	Beyond Breakpoints: A Tour Of Dynamic Analysis	QCon NYC Materials
12.2015	Two Approaches Towards OS Scalability	Papers We Love SF Video
09.2015	Racing to Win: Correct Concurrency with Race Conditions	Surge '15 Video
04.2015	Your Computer Is Already A Distributed System; Why Isn't Your OS?	Papers We Love SF Video
06.2014	Your Heap And You: Garbage Collector Tuning for Twitter Services	Internal tech talk
05.2013	Cachekata: Memory Hierarchy Optimization via Dynamic Binary Translation	Msc. Thesis PDF
04.2013	Whose Cache Line is it Anyway: OS Support for Live Detection & Repair of False Sharing	Eurosys '13 PDF
03.2012	Debugging Through Time with the Tralfamadore Debugger	RESolve '12 PDF
08.2011	Herbert West: Deanonymizer	HotSec'11 PDF
10.2010	Iodine: Interactive Program Partitioning	OSDI '10 Poster Session